FC7xxx FEE Integration Manual

Rev.A1

Revision History

Revision	Date	Changes
A0	2025/01/15	Initial release for MCAL
A1	2025/04/25	Update software implementation and integration manual need update too

Table of Contents

Chapter 1 Introduction

1.1 Introduction

This integration manual describes the integration requirements for the FEE module.

Chapter 2 Building

2.1 Dependencies on Other Modules

Module configuration dependency

- Fls: This module provides physical flash sectors for FEE module to manage memory
- Common: This module is the basic module which used to choose the chip.

Module build dependency

- Common: This module provides common type for other modules.
- Rte: This module provides APIs to protect/unprotect some parts of code from interrupts (Exclusive Areas).
- Det: This module is necessary for enabling Development error detection.
- Fls: This module provides specific flash operation APIs like write/read…
- MemIf: This module provides return type of memory operation.

Module initialization dependency

- Fls: FEE module is the upper layer module which works on Fls module, so Fls module must be initialized before it.

2.2 Files Required for Compile

FEE module files:

- MCAL/Src/Fee/Src/Fee_ChunkSwitch.c
- MCAL/Src/Fee/Src/Fee_Extra.c
- MCAL/Src/Fee/Src/Fee_Initialization.c
- MCAL/Src/Fee/Src/Fee_Internal.c
- MCAL/Src/Fee/Src/Fee_Reserve.c
- MCAL/Src/Fee/Src/Fee_RestoreJob.c
- MCAL/Src/Fee/Src/Fee.c
- MCAL/Src/Fee/include/Fee_Cbk.h
- MCAL/Src/Fee/include/Fee_ChunkSwitch.h
- MCAL/Src/Fee/include/Fee_Extra.h
- MCAL/Src/Fee/include/Fee_Initialization.h
- MCAL/Src/Fee/include/Fee_Internal.h
- MCAL/Src/Fee/include/Fee_InternalTypes.h
- MCAL/Src/Fee/include/Fee_MemoryMap.h
- MCAL/Src/Fee/include/Fee_Reserve.h
- MCAL/Src/Fee/include/Fee_RestoreJob.h
- MCAL/Src/Fee/include/Fee_Types.h
- MCAL/Src/Fee/include/Fee_version.h
- MCAL/Src/Fee/include/Fee.h
- MCAL_generate/src/Fee_Cfg.c
- MCAL_generate/include/Fee_Cfg.h

Fls module files:

- MCAL/Src/Fls /include/Fls.h

Common module files:

- MCAL/Src/Common/include/Mcal.h

- MCAL/Src/Common/include/Std_Types.h
- MCAL/Src/Common/include/Platform_Types.h
- MCAL/Src/Common/include/Compiler.h
- MCAL/Src/Common/include/CompilerDefinition.h

Det module files:

- Det.h

Rte module files:

- SchM_Fee.h

2.3 Add Plug-ins

FEE module plug-ins are developed for EB tresos Studio, so, to use FEE plug-ins on the EB tresos Studio, the user needs to add the FEE module to the EB plug-ins folder first.

1) Copy the FEE module (_MCAL/EB_Plugins/eclipse/plugins/Fee) folder to EB tresos plug-ins (EB/tresos/plugins/) folder.
2) Set the FEE module output location folder for the generated source file and header files.
3) Use EB tresos to configure the FEE module and generate source and header files.

Chapter 3 Memory

3.1 Sections in Memory Map

Section Name	Section Type	Description
FEE_START_SEC_VAR_NO_INIT_8 FEE_STOP_SEC_VAR_NO_INIT_8	Variables	These are all the sections used for variables which have to be aligned to 8 bit. These variables are never cleared and never initialized by startup code (bss).
FEE_START_SEC_CODE FEE_STOP_SEC_CODE	Code	Start and stop of memory Section for Code (text).
FEE_START_SEC_VAR_NO_INIT_ UNSPECIFIED FEE_STOP_SEC_VAR_NO_INIT_ UNSPECIFIED	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are never cleared and never initialized by startup code (bss).
FEE_START_SEC_VAR _INIT_ UNSPECIFIED FEE_STOP_SEC_VAR _INIT_ UNSPECIFIED	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are initialized by startup code(data).
FEE_START_SEC_CONST_UNSPECIFIED FEE_STOP_SEC_CONST_UNSPECIFIED	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit or boolean. These variables are read only(rodata).
FEE_START_CONFIG_DATA_UNSPECIFIED FEE_STOP_CONFIG_DATA_UNSPECIFIED	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit or boolean. These variables are read only(rodata).

Chapter 4 Exclusive Area

FEE module using the services of Schedule Manger (SchM) for entering and exiting critical regions.

The following critical regions are used in the FEE driver:

- Fee.c:
  - Fee_Read: exclusive area 0
  - Fee_Write: exclusive area 1
  - Fee_InvalidateBlock: exclusive area 2
  - Fee_EraseImmediateBlock: exclusive area 3

Chapter 5 Interrupt Service Routine (ISR)

None

Chapter 6 Error Report

6.1 Det

Function Name	Error Type
Fee_Init	FEE_E_UNINIT; FEE_E_INVALID_POINTER; FEE_E_BUSY_INTERNAL
Fee_SetMode	FEE_E_UNINIT; FEE_E_BUSY; FEE_E_BUSY_INTERNAL
Fee_Read	FEE_E_UNINIT; FEE_E_BUSY_INTERNAL; FEE_E_INVALID_BLOCK_NO; FEE_E_INVALID_BLOCK_OFS; FEE_E_INVALID_BLOCK_LEN; FEE_E_PARAM_POINTER
Fee_Write	FEE_E_UNINIT; FEE_E_INVALID_BLOCK_NO; FEE_E_PARAM_POINTER;
Fee_Cancel	FEE_E_UNINIT; FEE_E_INVALID_CANCEL
Fee_GetJobResult	FEE_E_UNINIT;
Fee_InvalidateBlock	FEE_E_UNINIT; FEE_E_BUSY; FEE_E_BUSY_INTERNAL; FEE_E_INVALID_BLOCK_NO
Fee_GetVersionInfo	FEE_E_PARAM_POINTER
Fee_EraseImmediateBlock	FEE_E_UNINIT;
Fee_MainFunction	FEE_E_UNINIT; FEE_E_PARAM_CHANNEL;
Fee_ForceChunkSwitchOnNextWrite	FEE_E_UNINIT;

6.2 Dem

None

Chapter 7 Function Calls

7.1 Function Calls during Startup

The API needs to be called is FEE_Init().

Note:

1) Fee module is the upper layer module which bases on Fls module.
2) Fls_Init() function must be called before calling the Fee_Init().
3) Fee_MainFunction() and Fls_MainFunction() must be called repeatedly for the FEE module initialization.

7.2 Function Calls during Shutdown

None

7.3 Function Calls during Wake-up

None

7.4 Function Calls during Runtime

Fee_MainFunction() and Fls_MainFunction() must be called repeatedly for the FEE module operation.

Chapter 8 Other Requirements

8.1 Notification, Callback, Callout

- Notification

  If user select FeeNvMJobEndNotification and FeeNvMJobErrorNotification on EB Tresos Studio configuration page, an extern declaration will generate in FEE_Cfg.h, and these two notifications are usually routed to the upper layer NvM.

- Callback

  FEE module publishes two APIs:

  Fee_JobEndNotification: This callback notification is used by the underlying Fls module to report the successful end of an Fls operation.

  Fee_JobErrorNotification: This callback notification is used by the underlying Fls module to report the failure of an Fls operation.

- Callout

  None

8.2 Macros

None

Chapter 9 Integration Steps

1) Configure the FEE module and generate configuration files (please refer to Building chapter for details).
2) Configure appropriate memory sections in linker file or other (please refer to Memory chapter for details).
3) Build the FEE module with dependent modules.